fn sample_negative_binomial_integer

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This document proves that the implementation of `sample_negative_binomial_integer` in `mod.rs` satisfies its proof definition.

1 Hoare Triple

Precondition

`shape` $\in \mathbb{N}_{>0}$ and `x` $\in \mathbb{Q}_{>0}$.

Pseudocode

```
1
2 def sample_negative_binomial_integer(shape, x) -> int:
3     total = 0
4     remaining = shape
5     while remaining != 0:
6         total += sample_geometric_exp_fast(x)  #
7         remaining -= 1
8     return total
```

Postcondition

For any setting of the input parameters `shape` and `x` such that the given preconditions hold, `sample_negative_binomial_inte` either returns `Err(e)` due to a lack of system entropy, or `Ok(out)`, where `out` is distributed as a negative binomial random variable on $\mathbb{N}_0$ with shape `shape` and success probability $1 - e^{-x}$, parameterized as the number of failures before the `shape`-th success.

2 Proof

Assume the preconditions are met.

Lemma 2.1. `sample_negative_binomial_integer` only returns `Err(e)` when there is a lack of system entropy.

Proof. Under the stated preconditions, the preconditions of `sample_geometric_exp_fast` on line 6 are met in every iteration of the loop. By its proof definition, each invocation of `sample_geometric_exp_fast` either returns a geometrically distributed sample or `Err(e)` due to a lack of system entropy. The function has no other fallible operations. Therefore any returned error must be due to a lack of system entropy. □

Lemma 2.2. If the outcome of `sample_negative_binomial_integer` is `Ok(out)`, then `out` is distributed as a negative binomial random variable on $\mathbb{N}_0$ with shape `shape` and success probability $1 - e^{-x}$, parameterized as the number of failures before the `shape`-th success.

Proof. Let m denote the value of `shape`. For each of the m iterations of the loop, line 6 draws an independent sample $G_i \sim Geometric(1 - e^{-x})$ on $\mathbb{N}_0$. The algorithm returns $K = \sum_{i=1}^{m} G_i$.

For any $k \in \mathbb{N}_0$, the event $\{K = k\}$ is equivalent to choosing a vector $(g_1, \ldots, g_m) \in \mathbb{N}_0^m$ such that $\sum_{i=1}^{m} g_i = k$. There are $\binom{k+m-1}{k}$ such vectors. Each vector has probability

$$\prod_{i=1}^{m} (1 - e^{-x}) e^{-x g_i} = (1 - e^{-x})^m e^{-xk}.$$

Therefore

$$\Pr[K = k] = \binom{k + m - 1}{k} (1 - e^{-x})^m e^{-xk}.$$

This is exactly the negative binomial distribution with shape m and success probability $1 - e^{-x}$, parameterized as the number of failures before the m-th success. □

Proof. 1 holds by Lemma 2.1 and Lemma 2.2. □